Msc Project: Theory and Methods for Certifying Deep Generative Models

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1 Introduction

1.1 Background and Motivation

Deep generative models aim to learn the underlying probability distribution of observed data and to generate new samples resembling those from the original data-generating process. Their ability to represent high-dimensional, complex distributions has driven substantial progress in areas such as image synthesis, data augmentation and scientific data generation. Among the principal families of deep generative models, denoising diffusion probabilistic models have attracted particular attention for their ability to generate high-quality samples through a sequence of learned denoising operations (Ho, Jain and Abbeel, 2020).

Despite this empirical success, reliably evaluating deep generative models remains challenging. Common approaches include visual inspection of generated samples and empirical measures such as the Fréchet Inception Distance (FID), which compares feature-space statistics of real and generated samples (Heusel et al., 2017). Like visual inspection, however, FID is a summary statistic computed from a finite batch of samples rather than a formal guarantee. In particular, finite-sample estimates of FID are biased, with the bias depending on both the sample size and the model being evaluated (Chong and Forsyth, 2020). it does not itself certify, with any stated confidence, how far the learned distribution lies from the unknown target distribution. Moreover, it depends on the chosen feature representation, the number of evaluation samples and the dataset considered. Strong empirical performance therefore does not necessarily establish that a generative model has accurately learned the target distribution.

This limitation motivates the study of certification for deep generative models. In this context, a certificate is a high-probability, data-dependent bound that quantifies the discrepancy between the target and learned distributions at a stated confidence level. Certification has been studied more extensively in supervised learning, where PAC-Bayesian methods have produced non-vacuous risk certificates for probabilistic neural networks (Pérez-Ortiz et al., 2021). Extending comparable guarantees to generative modelling is harder, however, since the object being learned is an entire distribution rather than a predictor evaluated through classification error. Recent work by Mbacke and Rivasplata (2024) makes progress towards this objective by deriving a quantitative upper bound on the Wasserstein distance between a data-generating distribution and the distribution learned by a diffusion model, providing a principled complement to sample-based and heuristic evaluation methods. Its practical value, however, depends on whether the bound can be computed reliably, whether it produces informative numerical values, and whether it extends beyond simple experimental settings. These questions motivate the present project: to reproduce the original certification experiment, evaluate the numerical behaviour of the resulting bound, and investigate its extension to further data distributions.

1.2 Research Gap and Problem Statement

PAC-Bayesian research has demonstrated that informative, non-vacuous certificates can be obtained for supervised neural networks, but these results concern prediction risk rather than the quality of a learned generative distribution (Pérez-Ortiz et al., 2021). For diffusion models, Mbacke and Rivasplata (2024) derived a finite-sample, high-probability upper bound on the Wasserstein distance between the target and learned distributions. However, the accompanying empirical evidence was limited to a two-dimensional uniform distribution, leaving the practical behaviour of the bound insufficiently explored.

Three limitations remain particularly relevant. First, it is unknown whether the reported numerical behaviour, including the non-vacuity and sensitivity of the bound to λ, extends to target distributions with different geometry or greater complexity. Second, the bound does not generally converge to zero as the sample size tends to infinity, indicating a potential limitation in the asymptotic tightness of the certificate, rather than necessarily of the diffusion model itself. Third, the numerical analysis estimates the required Lipschitz constants using the analytical quantity K_t', but the relationship between this estimate and the effective Lipschitz behaviour of the trained network is not empirically examined.. Consequently, the applicability and tightness of the certificate beyond the original synthetic setting remain open, including the question of which new datasets or target distributions provide appropriate and informative settings for further evaluation.

1.3 Certification and PAC-Bayesian Learning

The problem of certifying a learned model — producing a numerical guarantee, valid with a stated confidence, on a quantity that cannot be observed directly — has a long history in statistical learning theory. Early approaches bounded the gap between empirical and population risk using measures of hypothesis-class complexity, including VC dimension and Rademacher or Gaussian complexities (Vapnik, 1998; Bartlett and Mendelson, 2002). For modern neural networks these proved uninformative: Zhang et al. (2017) showed that standard architectures can fit arbitrary labellings of their training data, so any bound derived from class capacity alone must be extremely loose, and Neyshabur et al. (2017) reached similar conclusions across several candidate complexity measures. The resulting bounds were typically vacuous, exceeding the trivial worst-case value of the loss and therefore saying nothing about the trained model. Certification consequently shifted towards bounds that depend on the training data and on the specific predictor obtained, rather than on the size of the hypothesis class alone.

PAC-Bayesian analysis emerged as one of the most successful such approach, certifying a distribution over hypotheses by bounding its population risk through empirical risk and a Kullback–Leibler divergence from a prior fixed independently of the certification data (Shawe-Taylor and Williamson, 1997; McAllester, 1999), with subsequent refinements tightening the dependence on sample size (Maurer, 2004). A major empirical breakthrough came from Dziugaite and Roy (2017), who obtained the first non-vacuous bounds for stochastic networks with far more parameters than training examples by optimising the bound itself rather than evaluating it post hoc. Zhou et al. (2019) extended non-vacuous guarantees to ImageNet-scale architectures via compression. Pérez-Ortiz et al. (2021) then shifted attention from validity to tightness, training probabilistic networks on objectives derived from tight PAC-Bayes bounds and reporting certificates close to the observed test error — a distinction that matters because a bound may be correct yet too loose to be useful.

These results establish that informative numerical certification is achievable in supervised deep learning, and that the usefulness of a certificate depends not only on its formal validity but on whether its numerical value is small enough to be informative — a property that had to be pursued deliberately rather than obtained for free. Transferring this to generative modelling is not immediate, because the certified quantity is a discrepancy between two distributions rather than a classification risk, and the model is a multi-step stochastic sampler rather than a predictor. Mbacke, Clerc and Germain (2023) took a first step by deriving PAC-Bayesian guarantees for variational autoencoders, expressing the generation term as a Wasserstein distance; Mbacke and Rivasplata (2024) subsequently adapted this construction to DDPMs. For diffusion models, however, the corresponding numerical question remains largely unexamined: a valid quantitative bound now exists, but its behaviour across settings, and the conditions under which it stays informative, have received far less scrutiny than their supervised counterparts.

1.4 Distributional Guarantees for Diffusion Models

A central question in certifying a generative model is therefore which discrepancy should be used to compare the target and learned distributions. Common choices include Kullback–Leibler divergence, total variation distance and Wasserstein distance. These quantities capture different notions of distributional similarity. KL divergence may become infinite when the required absolute-continuity condition is violated, while total variation distance does not account for the geometry of the sample space. Wasserstein distance is instead defined through an optimal transport cost and, under appropriate moment conditions, can remain finite and informative even when the distributions have disjoint supports (Villani, 2009). This makes it particularly relevant to the evaluation of generative models, although no single discrepancy is universally preferable.

Much of the theoretical literature on diffusion models has analysed convergence through score-based or stochastic-differential-equation formulations. Song et al. (2021) introduced a continuous-time SDE framework that unified score-based generative models and diffusion probabilistic models. Subsequent work established convergence guarantees under assumptions involving score-estimation accuracy, sampling discretisation and properties of the target distribution. For example, Lee, Lu and Tan (2023) obtained Wasserstein convergence guarantees for distributions with bounded support or sufficiently decaying tails under an assumption of L^2-accurate score estimation. Although these results provide important theoretical understanding, assumptions on score-estimation error can make the resulting guarantees difficult to evaluate numerically for a particular trained model.

Mbacke and Rivasplata (2024) developed a different route by deriving a finite-sample, high-probability upper bound on the Wasserstein distance between the target distribution and the distribution learned by a denoising diffusion probabilistic model. Their result applies to arbitrary distributions on bounded instance spaces, does not require the target distribution to admit a density with respect to Lebesgue measure, and avoids assumptions on the learned score function. However, the reported numerical illustration focused on a simple two-dimensional uniform distribution. Consequently, the numerical behaviour, tightness and practical informativeness of the bound in broader experimental settings remain insufficiently understood. These considerations underlie the limitations identified in Section 1.2.

1.5 Aim, Objectives and Research Questions

To achieve this aim, the project has the following objectives: To reproduce the original two-dimensional diffusion-model experiment and compare the resulting Wasserstein upper bounds with the values reported by Mbacke and Rivasplata (2024).

To investigate how the trade-off parameter λ affects the numerical value and non-vacuity of the certificate.

To identify an appropriate new dataset or target distribution for extending the original experiment, considering properties such as bounded support, dimensionality and data geometry.

To make the model and evaluation modifications required by the selected dataset and assess whether the resulting certificate remains numerically informative beyond the original synthetic setting.

To provide reproducible code, experimental results and a research report documenting the methodology, findings and limitations of the investigation.

These objectives are organised around three research questions:

RQ1: To what extent can the Wasserstein upper-bound values reported for the original two-dimensional experiment be reproduced?

RQ2: How does the trade-off parameter λ affect the numerical behaviour and non-vacuity of the Wasserstein upper bound?

RQ3: To what extent can the certification method be extended to a new dataset or target distribution while retaining an informative numerical upper bound?

1.6 Contributions

This project makes the following empirical and methodological contributions to the evaluation of Wasserstein-based certificates for diffusion models: It provides a systematic replication of the numerical experiment reported by Mbacke and Rivasplata (2024), including all six values of the trade-off parameter λ, and quantitatively compares the reproduced bounds with the published results.

It examines the effect of λ on the numerical value and non-vacuity of the certificate.

It develops a theoretically informed procedure for selecting an additional dataset or target distribution, accounting for bounded support, dimensionality and data geometry, and adapts the model and certificate evaluation accordingly.

It provides a reproducible experimental implementation that separates model training from certificate evaluation, enabling multiple bound settings to be assessed using the same trained diffusion model.

2 Methodology

2.1 Theoretical Framework

2.1.1 Denoising Diffusion Probabilistic Models

The generative model used in this project is a discrete-time DDPM, based on the diffusion framework of Sohl-Dickstein et al. (2015) and the parameterisation of Ho, Jain and Abbeel (2020). A DDPM can also be interpreted as a Markovian hierarchical variational autoencoder in which the encoder is represented by the fixed forward diffusion process and the decoder corresponds to the learned multi-step reverse process (Luo, 2022). This interpretation provides the methodological connection between the two principal works underlying the present project. Mbacke, Clerc and Germain (2023) first developed PAC-Bayesian reconstruction and Wasserstein generation guarantees for variational autoencoders. Mbacke and Rivasplata (2024) subsequently adapted this framework to DDPMs by treating the terminal forward distribution $q(x_T \mid x_0)$ as the conditional latent distribution and the Gaussian distribution $p(x_T)$ as the latent prior. Let $x_0 \sim \mu$ denote a sample from the target distribution, where $x_0 \in R^D$. The forward process is a Markov chain with transitions

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{\alpha_t} x_{t-1}, (1-\alpha_t)I\right),$$

where $t \in \{1, \ldots, T\}$, $\alpha_t = 1-\beta_t$, and β_t controls the amount of noise introduced at time step t. Defining

$$\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s,$$

the distribution of x_t conditioned directly on x_0 has the closed form

$$q(x_t \mid x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)I\right).$$

Consequently, a noisy sample can be generated without simulating all preceding time steps:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \qquad \epsilon \sim \mathcal{N}(0, I).$$

For a sufficiently long and appropriately selected noise schedule, the terminal distribution $q(x_T \mid x_0)$ becomes close to the standard Gaussian prior $p(x_T) = \mathcal{N}(0, I)$. The learned reverse process attempts to invert this transformation. Its transitions are represented as

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\left(x_{t-1}; g_\theta^t(x_t), \sigma_t^2 I\right),$$

where g_θ^t is a neural-network-dependent mean function and σ_t^2 is determined by the variance schedule. Following Ho, Jain and Abbeel (2020), the neural network can be parameterised to predict the Gaussian noise ϵ that produced x_t. During training, a data sample, a time step and a noise vector are sampled, after which the model minimises the simplified noise-prediction objective

$$\mathcal{L}_{\text{DDPM}}(\theta) = E_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right].$$

After training, generation begins with $x_T \sim \mathcal{N}(0, I)$ and repeatedly applies the learned reverse transitions until an approximate data sample x_0 is obtained. All subsequent experiments and certificate calculations therefore use this discrete-time DDPM formulation. In the replicated two-dimensional experiment, the noise-prediction function is implemented using a fully connected neural network conditioned on sinusoidal embeddings of the noisy coordinates and the diffusion time step; the detailed architecture and training configuration are presented in Section 2.2.

2.1.2 Wasserstein Distance

In this project, the discrepancy between the target distribution μ and the distribution learned by the model π_θ is quantified using the first-order Wasserstein distance $W_1(\mu, \pi_\theta)$. This metric is adopted because it incorporates the geometry of the underlying data space, so the discrepancy depends not only on differences in probability mass but also on the distance over which that mass must be transported (Villani, 2009; Arjovsky, Chintala and Bottou, 2017).

Let p and q be probability distributions on a metric space $\mathcal{X}$. A coupling γ of p and q is a joint distribution on $\mathcal{X} \times \mathcal{X}$ whose marginal distributions are p and q, respectively. The set of all such couplings is denoted by $\Gamma(p,q)$. For distributions on R^D equipped with the Euclidean metric, the Wasserstein distance of order one is defined as

$$W_1(p,q) = \inf_{\gamma\in\Gamma(p,q)} \int_{\mathcal{X}\times\mathcal{X}} \|x-y\|_2 \, d\gamma(x,y).$$

Intuitively, $W_1(p,q)$ represents the minimum expected transportation cost required to transform probability mass distributed according to p into probability mass distributed according to q, where the cost of transporting mass from x to y is their Euclidean distance. Unlike total variation distance, the Wasserstein distance accounts for how far apart samples are in the underlying space. It can also remain finite and informative when two distributions have disjoint supports, whereas the Kullback–Leibler divergence may be infinite when the required absolute-continuity condition is not satisfied (Villani, 2009; Mbacke and Rivasplata, 2024).

The Wasserstein distance is a metric and therefore satisfies the triangle inequality. This property is central to the certification method used in this project. Denoting the target distribution by μ, the distribution learned by the diffusion model by π_θ, and a sample-dependent regenerated distribution by μ_θ^n, the target discrepancy can be decomposed as

$$W_1(\mu,\pi_\theta) \le W_1(\mu,\mu_\theta^n) + W_1(\mu_\theta^n,\pi_\theta).$$

The two terms can then be controlled separately using the empirical reconstruction loss, prior matching and the Lipschitz properties of the reverse diffusion process. The boundedness of the instance space also provides a baseline for interpreting the resulting certificate. If both distributions are supported on a space with finite diameter

$$\Delta = \sup_{x,x'\in\mathcal{X}} \|x-x'\|_2,$$

then the product coupling gives $W_1(p,q) \le \Delta$. In the replicated experiment, $\mathcal{X} = [-1,1]^2$, and hence

$$\Delta = \sqrt{(2)^2+(2)^2} = \sqrt{8} \approx 2.828.$$

A computed upper bound below $\sqrt{8}$ therefore improves upon this direct bounded-space baseline and is treated as non-vacuous in the original numerical experiment.

2.1.3 Lipschitz Continuity

The Wasserstein certificate used in this project requires each reverse-process mean function to satisfy a Lipschitz continuity condition (Mbacke and Rivasplata, 2024). Let $g : \mathcal{X} \to \mathcal{Y}$ be a function between Euclidean spaces. The function is K-Lipschitz continuous if there exists a finite constant $K \ge 0$ such that

$$\|g(x)-g(y)\|_2 \le K\|x-y\|_2 \qquad forall x, y \in \mathcal{X}.$$

The smallest constant satisfying this condition is called the Lipschitz constant of g. It bounds the maximum extent to which a difference between two inputs can be amplified by the function; in particular, $K < 1$ implies that the function is contractive (Gouk et al., 2021). For the reverse diffusion process, Mbacke and Rivasplata (2024) assume that the mean function g_θ^t at every time step is K_θ^t-Lipschitz continuous:

$$\left\|g_\theta^t(x) - g_\theta^t(y)\right\|_2 \leq K_\theta^t \|x - y\|_2.$$

This assumption is required because a diffusion model applies a sequence of reverse transitions rather than a single mapping. The Lipschitz constant of a composition is bounded by the product of the constants of its component functions (Gouk et al., 2021). Consequently, a difference introduced at the terminal diffusion state may propagate through the complete reverse process according to products of the form

$$\prod_{t=1}^{T} K_\theta^t.$$

Following the Gaussian parameterisation of the DDPM reverse process, a transition at $t \geq 2$ can be represented as

$$x_{t-1} = g_\theta^t(x_t) + \sigma_t \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$$

(Ho, Jain and Abbeel, 2020). For two independently sampled reverse transitions, the Lipschitz condition and the triangle inequality give

$$E\|x_{t-1} - y_{t-1}\|_2 \leq K_\theta^t \|x_t - y_t\|_2 + \sigma_t E\|\epsilon - \epsilon'\|_2$$

(Mbacke and Rivasplata, 2024, Lemma 3.3). Repeated application of this inequality produces two components of the final certificate. The product of the Lipschitz constants controls how the discrepancy between the terminal forward distribution $q(x_T \mid x_0)$ and the Gaussian prior $p(x_T)$ propagates through the reverse process. A second, weighted sum controls the accumulation of Gaussian sampling noise across the reverse steps (Mbacke and Rivasplata, 2024, Lemmas 3.4–3.5). The true constants K_θ^t of the trained neural network are not calculated directly in the original numerical experiment. Instead, Mbacke and Rivasplata (2024, Remark 3.2) use the Lipschitz constant of the corresponding forward posterior mean as a proxy:

$$K_t' = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t}.$$

For $0 < \alpha_t < 1$, this quantity satisfies $K_t' < 1$. Its use is motivated by the fact that the learned reverse mean is trained to approximate the forward posterior mean. Nevertheless, K_t' is a schedule-derived proxy rather than a measured Lipschitz constant of the trained network. The same approximation is retained when computing the Lipschitz-dependent terms in the replicated certificate.

2.1.4 Regenerated Distribution and Reconstruction Loss

This project uses the regenerated distribution to connect reconstruction from observed data with generation from the diffusion prior. For a sample x_0, define

$$\pi_\theta(\cdot \mid x_0) = q(x_T \mid x_0) p_\theta(x_{T-1} \mid x_T) \cdots p_\theta(\cdot \mid x_1).$$

This is the distribution obtained by first sampling x_T from the terminal forward distribution conditioned on x_0, and then passing it through the complete learned reverse process. Given an i.i.d. sample $S = \{x_0^1, \ldots, x_0^n\}$, the regenerated distribution is

$$\mu_\theta^n = \frac{1}{n} \sum_{i=1}^{n} \pi_\theta(\cdot \mid x_0^i).$$

Unlike the learned distribution π_θ, which begins from the Gaussian prior $p(x_T)$, μ_θ^n begins from noisy versions of observed samples. The associated reconstruction loss is

$$\ell_\theta(x_T, x_0) = E_{p_\theta(\hat{x}_0 \mid x_T)} \left[x_0 - \hat{x}_{02} \right],$$

where the expectation represents the complete reverse process.

2.1.5 PAC-Bayesian-Style Analysis

This project follows the PAC-Bayesian-style analysis used by Mbacke and Rivasplata (2024) to control the distance between the target distribution μ and the regenerated distribution μ_θ^n. In this construction, the data-dependent terminal distribution $q(x_T \mid x_0)$ plays a role analogous to a posterior distribution, while the Gaussian distribution $p(x_T)$ acts as a data-independent prior. Their discrepancy is therefore measured by

$$\mathrm{KL}(q(x_T \mid x_0) \,\|\, p(x_T)).$$

Combining this prior-matching term with the empirical reconstruction loss gives, with probability at least $1 - \delta$,

$$W_1(\mu, \mu_\theta^n) \leq \frac{1}{n} \sum_{i=1}^{n} E_{q(x_T \mid x_0^i)} \ell_\theta(x_T, x_0^i) + \frac{1}{\lambda} \left[\sum_{i=1}^{n} \mathrm{KL}\big(q(x_T \mid x_0^i) \| p(x_T)\big) + \log \frac{1}{\delta} \right] + \frac{\lambda \Delta^2}{8n}.$$

Unlike conventional PAC-Bayes analysis, this construction concerns distributions over terminal latent variables rather than distributions over network weights.

2.1.6 PAC-Bayesian-Style Wasserstein Bound

This project uses the finite-sample Wasserstein certificate derived by Mbacke and Rivasplata (2024). If $\mathcal{X}$ has finite diameter Δ, then for $\lambda > 0$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$W_1(\mu, \pi_\theta) \leq \frac{1}{n} \sum_{i=1}^{n} E_{q(x_T \mid x_0^i)} \ell_\theta(x_T, x_0^i) + \frac{1}{\lambda} \left[\sum_{i=1}^{n} \mathrm{KL}\big(q(x_T \mid x_0^i) \,\|\, p(x_T)\big) + \log \frac{1}{\delta} \right] + \frac{\lambda \Delta^2}{8n} + \left(\prod_{t=1}^{T} K_\theta^t \right) \frac{1}{n} \sum_{i=1}^{n} E_{q(x_T|}$$

The terms respectively represent reconstruction error, prior matching and confidence, the bounded-space penalty, propagated terminal mismatch, and accumulated reverse-process noise. This project computes their sum as the estimated numerical certificate for $W_1(\mu, \pi_\theta)$.

2.2 Replication of the Original Experiment

2.2.1 Code Provenance and Scope of the Replication

The replication used the authors' supplementary implementation for Mbacke and Rivasplata (2024), obtained from the paper's OpenReview submission. The original data generation, diffusion model, training procedure and Wasserstein certificate calculations were retained unchanged. Modifications were limited to experiment management, including model saving/loading, automated result recording and evaluation of multiple λ values using the same trained model. The aim was therefore to reproduce the published numerical results using the supplied implementation rather than to reimplement or modify the method.

2.2.2 Experimental Data

The target distribution was the uniform distribution on a two-dimensional square of side length two centred at the origin:

$$\mu = \mathcal{U}([-1, 1]^2).$$

The data were generated synthetically by sampling the two coordinates independently from $\mathcal{U}(-1, 1)$. The resulting instance space was $\mathcal{X} = [-1, 1]^2$, with Euclidean diameter

$$\Delta = \sqrt{(2)^2 + (2)^2} = \sqrt{8}.$$

A training set containing 50,000 samples was generated for model optimisation. A separate set of $n = 5,000$ samples was generated after training and used exclusively for certificate evaluation. A further 2,000 target samples and 2,000 model-generated samples were used for qualitative visualisation but did not contribute to the certificate calculation.

2.2.3 Model Architecture

The generative model was a discrete-time DDPM with $T = 1,000$ diffusion steps. A linear variance schedule was used, increasing from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. A single shared noise-prediction network was applied at every diffusion step. Its inputs were the two coordinates of the noisy sample, $x_{t,1}$ and $x_{t,2}$, and the diffusion time step t. Each scalar input was mapped independently to a 128-dimensional sinusoidal embedding. The three embeddings were concatenated to produce a 384-dimensional input vector. The network consisted of four fully connected hidden layers. The first layer mapped the 384-dimensional input to 128 hidden units, followed by three $128 \to 128$ layers. Each hidden layer used a GELU activation. The final $128 \to 2$ linear layer predicted the two-dimensional Gaussian noise vector. The network contained 99,074 trainable parameters

2.2.4 Training Procedure

The model was trained using 50,000 synthetically generated samples. The training `DataLoader` was configured with a batch size of 100, and optimisation was performed for 500 epochs. For each training example, a diffusion time step t and a Gaussian noise vector $\epsilon \sim \mathcal{N}(0, I)$ were sampled independently, and the noisy observation x_t was generated directly from the corresponding clean sample using the precomputed diffusion schedule.

The network was trained to predict the sampled Gaussian noise from (x_t, t). The training objective was the mean-squared error between the true and predicted noise. Optimisation used AdamW with a learning rate of 10^{-4}, with the remaining optimiser settings left at their PyTorch defaults.

After training, the parameters of the reverse-process network were saved to disk. This trained model was subsequently reused for certificate evaluation, so no retraining was performed when evaluating different values of λ.

No explicit random seed was set in either the training or the evaluation script. In addition, the synthetic datasets were generated using NumPy's `default_rng()` without a specified seed, so a fresh certification set was drawn each time the evaluation script was run. The replication was therefore stochastic rather than fully deterministic.

2.2.5 Certificate Evaluation Procedure

The bound was evaluated separately from training, with the loaded model parameters held fixed, on a new independent certification set of $n = 5{,}000$ samples, using $\delta = 0.05$ and $\Delta = \sqrt{8}$. The reconstruction component was estimated using one terminal draw from $q(x_T \mid x_0)$ and one stochastic reverse trajectory per certification sample, while the prior-matching term was evaluated analytically. The Lipschitz quantities were computed using the schedule-derived expression implemented in the authors' supplied code, with the first value set to unity as specified in the implementation. The expected Gaussian-distance terms were estimated by Monte Carlo using 10^6 draws.

The reconstruction, KL sum, propagated terminal-mismatch and accumulated-noise quantities were computed once and reused across $\lambda \in \{n/10,\ n/5,\ n/2,\ n,\ n/0.5,\ n/0.1\}$. Across these evaluations, only the two PAC-Bayesian terms involving λ, namely $(\mathrm{KL} + \log(1/\delta))/\lambda$ and $\lambda\Delta^2/(8n)$, changed. Theorem 3.1 provides a high-probability guarantee for each fixed λ rather than a simultaneous guarantee over all six values; the resulting values were therefore treated as separate certificates.

2.2.6 Replication Setting

A comparison between the theoretical description and the released implementation identified several implementation-level differences. First, although the training DataLoader is configured with a batch size of 100, the supplied training loop applies `batch[0]` before computing the loss, so the effective update does not use the full minibatch as configured. Second, in the accumulated reverse-process noise term, σ_t is multiplied within the product over preceding Lipschitz factors rather than outside it, and is implemented using the forward variance β_t. Third, the implemented Lipschitz expression uses $\sqrt{\bar{\alpha}_t}$, whereas Remark 3.2 specifies $\sqrt{\alpha_t}$. These behaviours were retained unchanged because the aim was to reproduce the released implementation rather than construct a corrected implementation of the theoretical bound.

2.3 Extension

3 Result and Evaluation

3.1 Replication of the Original Experiment

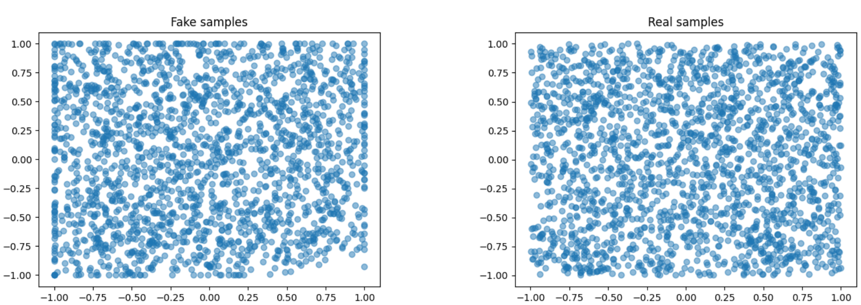
3.1.1 Replication 1

3.1.1.1 Reproduction of the Published Bound Values

Table 1: Comparison between published and replicated results

λ	Published	Replication	Absolute Difference	Relative Difference
$n/10$	1.124	1.1535	0.0295	2.62%
$n/5$	1.231	1.2505	0.0195	1.58%
$n/2$	1.518	1.5486	0.0306	2.02%
n	2.035	2.0480	0.0130	0.64%
$n/0.5$	3.056	3.0477	0.0083	0.27%
$n/0.1$	11.061	11.0475	0.0135	0.12%

3.1.1.2 Comparison of Generated Samples



(a) Generated samples (b) Real samples

Figure 1: Comparison between generated and real samples

4 Conclusion